Diploma in Information Technology
Final Practical Assignment

Submission Type: Printed Book / Report
Individual Assignment

Students must select ONLY ONE task from the following two tasks.

General Instructions

1. Select only one assignment task.
2. The assignment must be completed individually.
3. Submit the assignment as a printed report/book.
4. The report must include Cover Page, Student Name, Student ID, Course Name, Assignment Title, Introduction, Objectives, Methodology, Screenshots/Diagrams, Source Code, Output Results, and Conclusion.
5. All screenshots must be clear and properly labeled.
6. Plagiarism is strictly prohibited.
7. Late submissions may receive marks deductions.

Task 1 – Python Calculator Application

Assignment Title: Development of a Basic Calculator Using Python

Objective:
Develop a calculator program using Python that can perform basic mathematical operations.

Requirements:
The calculator must perform: Addition Subtraction Multiplication Division **Inputs:**
- Number 1 (no1)
- Number 2 (no2)
- Operation (+, -, *, /)

Functional Requirements: Ask the user to enter two numbers. Ask the user to enter the required operation. Perform the calculation. Display the result correctly. Handle division carefully (avoid division by zero).

```
Enter first number: 10
Enter second number: 5
Enter operation (+,-,*,/): *

Result = 50
```

Submission Requirements:
- Python source code
- Screenshots of program execution
- Explanation of the code
- Output examples

Task 2 – Arduino Automatic Watering System

Assignment Title: Development of an Automatic Watering System Using Arduino

Objective:
Design and simulate an automatic watering system using Arduino and a soil moisture sensor.

Required Components: Arduino Uno Soil Moisture Sensor Water Pump / LED (Simulation) Breadboard Jumper Wires **Simulation Platform:**
Tinkercad.com

Functional Requirements: Read soil moisture values. Detect dry soil conditions. Automatically activate watering when soil is dry. Stop watering when moisture level becomes sufficient. **Submission Requirements:**
- Hardware connection diagram
- Arduino source code
- Tinkercad simulation screenshots
- Explanation of system functionality
- Output screenshots/results

Recommended Report Structure

1. Cover Page
2. Declaration
3. Acknowledgement
4. Table of Contents
5. Introduction
6. Objectives
7. Materials / Software Used
8. Methodology
9. Implementation
10. Screenshots / Diagrams
11. Source Code
12. Results and Testing
13. Conclusion
14. References

Marking Scheme

Criteria	Marks
Correct Functionality	30
Code Quality	20
Report Quality	15
Screenshots / Diagrams	15
Creativity & Presentation	10
Viva / Demonstration	10
Total	**100**

Submission Deadline:
Lecturer will announce the submission date separately.

Important Notes:
- Students must ensure the assignment is fully tested before submission.

- Printed reports should be properly bound.
- Source code must be included in the report appendix.
- Students may be asked to demonstrate their project during evaluation.